Predicting Water System Violations

Water is essential for all life, but not all water is clean and drinkable. There are many cases and many concerns about the cleanliness of water, whether it’s truly pure or contaminated with things we cannot see like bacteria and viruses. A last year Article *Toxic Tap Water* by Evangelyn Rodriguez for EPA WATCH.org explains that there are many hidden chemicals and heavy metals in tap water that can be linked to things like cancer. The concern for this is real.

On a grander scale are water system violations. Huge supplies of water that exceed contamination limits, people failing to report test results, pipe bursting’s, etc. There are many reasons why this can happen and it can easily result in ongoing public health issues, especially in low-income areas and communities. As a group we want to come together and see if we can predict which water systems are likely to get violations (whatever the cause) before they happen and be able to create visualizations detailing high risk areas ahead of time. Ultimately, we want to know:

Can we predict water system violations based on drinking water data and use geographic mapping to show high-risk areas?

To answer this question, we are using the EPA safe drinking water information system that we accessed through the EPA ECHO data downloads:
(https://catalog.data.gov/dataset/safe-drinking-water-information-system-sdwis)

This dataset covers around 160,000 public water systems all around the U.S. and it included monitoring, enforcement, and violation data that was collected by states and then reported to the EPA. The main features of this data set we plan to use include categories of state (geographical mapping), water source type (possible difference in contamination risk between surface and ground water), historical violation counts (likely strongest predictor), and population served (so we can see if smaller systems tend to have fewer compliance resources.

We began by importing all necessary libraires. Pandas for tables/dataframes, seaborn and matplotlib for visualization, NumPy for all necessary numerical calculations, and Scikit-learn as our machine learning library of choice. We loaded all the data in, using “popmean” to contain the mean annual population served by country from 1994-2016.

We decided to use FIPS code which are numeric identifiers for counties in the United States. A small but addressed matter is that numeric identifiers may start with zero like “01001” corresponding to Autauga County, Alabama. Python may drop the leading zero leading to it being read as “1001” which is a different code. We then created a function to clean the data and make sure it’s ready for merging. To fix the problem stated above we converted FIPS columns to text and padded the numbers, adding 0s to the top until they were all five digits long. The cleaned result is stored in a column named ‘FIPS’. Then we applied this to the other data sets, cleaning the ‘FIPS’ column in every dataset to merge them later. This was done now because without standardized FIPS codes merging data could fail or produce missing values.

When we were ready to merge, we created a separate variable df_final with .copy() to ensure that changes made to this variable wouldn’t affect the original variables. With df_final we merged the violations dataframe keeping only FIPS and mean_viol_cnty, added population served keeping only FIPS and pop_served and then merge gw which contains FIPS codes, groundwater percentage violations, and finally surface water violations including FIPS codes and percentage. Afterwards we made sure everything worked by checking the head of the data and whether it had any missing value.

	State_County	State	STATE_FIPS	CNTY_FIPS	COUNTY	COUNTY2	FIPS	FIPS2	pct_gw_violations_x	mean_viol_cnty	Pop_Served
0	AK_Aleutians East Borough	AK	2	13	Aleutians East Borough	Aleutians East Borough	02013	f02013	0.000000	NaN	0
1	AK_Aleutians West Census Area	AK	2	16	Aleutians West Census Area	Aleutians West Census Area	02016	f02016	0.621118	0.043478	697
2	AK_Anchorage Municipality	AK	2	20	Anchorage Municipality	Anchorage Municipality	02020	f02020	0.071667	0.130435	534
3	AK_Bethel Census Area	AK	2	50	Bethel Census Area	Bethel Census Area	02050	f02050	0.000000	NaN	0
4	AK_Bristol Bay Borough	AK	2	60	Bristol Bay Borough	Bristol Bay Borough	02060	f02060	0.000000	NaN	0

When merging everything to df_final we merged it with gw and all its data came over but actually since df_final is a copy of the original dataset that already included df it was uneccsary and would actually cause duplicate data. To prevent mistakes down the line we removed it here. After we filled any missing values with zero for modeling. We also created a new column violations per 1000 people. This is because a county serving a million people would naturally have more violations than a county only serving a couple of thousand due to a larger water system. Instead of using raw violation count the code counts violations per 1000 people which acts as a better target variable which is what we created next. Using the target variable the model's goal was to predict whether a country is above or below the median violation rate. After checking our data types, we began to create a machine learning model.

The model we decided to use was a Gradient Boosting classifier, a model capable of making many small decision trees one after another. Each new tree tries to fix mistakes made by the previous trees. We used the variables population served (pop_served), percentage of groundwater systems with nitrate violations (pct_gw_violations), and the percentage of surface water systems with nitrate violations. With this the model tries to predict a high violation (high_violation'). We split the dataset into four so we could test and train the model. We created the model and stored it in classifer (clf) and trained the model on the test dataset. The mode's job is to study relationships between the features and the targets, like ifvcounties with higher groundwater violation percentages tend to have high overall violation rates or if population served influence the prediction. We then scored the model to check for accuracy, and it gave .8352 or 83.52% accuracy. This means the model was able to accurately able to predict whether a country was 'high violation' or 'not high violation' about 83% of the time. Using feature importance to measure how important each variable was to the model we discovered that the model mostly relied heavily on population served to make its predictions. This is important because it suggests that county water system size is strongly related to whether the county is classified as high violation. Much less influential than expected were groundwater violation percentage and surface water violations, the latter of which barely mattered at all. Either was 83% is a strong score for a basic model. At that percentage it's best to say that the model can usually predict, instead of mostly. Now accuracy isn't everything. Other factors like precision, recall, even an F1-score might be useful to check out. 83% while a great number does not show what kind of mistakes the model made if it did. Even still it is a great number to start with as it shows the potential ability for future models, better ones, to produce even greater results.